EXPERIMENT 5

Electromagnetic Simulation and S-Parameter Characterization of a Microwave Directional Coupler and Two-Way Power Divider

1. Aim

To analyze the S-parameter response of a microwave directional coupler and a two-way power divider over 8–12 GHz, and to evaluate coupling, isolation, insertion loss, directivity and equal power division at a design frequency of 10 GHz.

2. Objectives

1. To design microwave directional-coupler and power-divider structures for a specified operating frequency.
2. To determine coupling factor, isolation, insertion loss and directivity of the directional coupler.
3. To analyze power division ratio, port matching and output-port isolation of the power divider.
4. To optimize the physical dimensions for improved S-parameter performance over the desired operating bandwidth.

3. Parameters Used

Parameter	**Value**
Design frequency	10 GHz
Reference impedance	50 Ω
Frequency sweep	8–12 GHz
Number of frequency points	401
Coupling target	10 dB
Isolation target	> 25 dB
Insertion-loss target	< 1.5 dB
Directivity target	≥ 15 dB
Length change for comparison	5%
Ideal divider output levels	−3.01 dB each

4. MATLAB Code

Save the following script as Experiment5.m and run it in MATLAB. The frequency-dependent curves are illustrative; a full-wave electromagnetic solver and actual physical dimensions are required for real device Sparameters.

```
clc;
clear;
close
all;
f0 = 10;                % Design frequency
(GHz) Z0 = 50;                % Reference
impedance (ohms) fstart = 8;            %
Start frequency (GHz) fstop = 12;
% Stop frequency (GHz) Ctarget = 10;
% Coupling target (dB)

f = linspace(fstart, fstop,
401); S11 = -25 - 3*((f -
f0)/2).^2;
S21 = -0.5 - 0.1*(f - f0).^2;
S31 = -Ctarget - 0.5*(f -
f0).^2;
S41 = -28 - 2*cos((f -
f0)*pi/2);

Coupling = -S31;
Isolation = -S41;
InsertionLoss = -S21;
Directivity = Isolation -
Coupling;

S11_div = -25 - 2*((f -
f0)/2).^2;
```

```
S21_div = -3.01 - 0.1*(f -
f0).^2;
S31_div = -3.01 - 0.1*(f -
f0).^2;
S23_div = -30*ones(size(f));

P21 = 10.^(S21_div/10);
P31 = 10.^(S31_div/10);

PowerRatio = P21 ./ P31;

[~, idx] = min(abs(f - f0));

fprintf('\nEXPERIMENT 5 RESULTS AT %.2f
GHz\n', f(idx)); fprintf('Reference impedance = %.0f
ohms\n', Z0); fprintf('Frequency sweep = %.1f to
%.1f GHz\n', fstart, fstop);

fprintf('\nDIRECTIONAL COUPLER\n'); fprintf('S11 = %.2f
dB\n', S11(idx)); fprintf('S21 = %.2f dB\n', S21(idx));
fprintf('S31 = %.2f dB\n', S31(idx)); fprintf('S41 (isolated
port) = %.2f dB\n', S41(idx)); fprintf('Coupling = %.2f dB
(target 10 dB)\n', Coupling(idx)); fprintf('Isolation = %.2f dB
(target >= 25 dB)\n', Isolation(idx)); fprintf('Insertion loss =
%.2f dB (target <= 1.5 dB)\n', InsertionLoss(idx));
fprintf('Directivity = %.2f dB (target >= 15 dB)\n',
Directivity(idx));

fprintf('\nTWO-WAY POWER DIVIDER\n'); fprintf('S11
= %.2f dB\n', S11_div(idx)); fprintf('S21 = %.2f dB (target
-3.01 dB)\n', S21_div(idx)); fprintf('S31 = %.2f dB (target
-3.01 dB)\n', S31_div(idx)); fprintf('Output isolation S23 =
%.2f dB\n', S23_div(idx)); fprintf('Power division ratio
```

```
P21/P31 = %.3f:1\n', PowerRatio(idx)); figure('Name',
'Experiment 5 - Directional Coupler and Power Divider', ...
    'NumberTitle', 'off', 'Color', 'w') subplot(3, 4, 1); plot(f, S11, 'LineWidth', 1.5); grid on; title('Coupler
S11'); xlabel('Frequency (GHz)'); ylabel('Magnitude (dB)');

subplot(3, 4, 2); plot(f,
S21, 'LineWidth', 1.5);
grid on; title('Coupler
S21'); xlabel('Frequency
(GHz)');
ylabel('Magnitude
(dB)');

subplot(3, 4, 3); plot(f,
S31, 'LineWidth', 1.5);
grid on; title('Coupler
S31'); xlabel('Frequency
(GHz)');
ylabel('Magnitude
(dB)');

subplot(3, 4, 4);
plot(f, S41, 'LineWidth', 1.5);
grid on; title('Coupler S41 -
Isolated Port');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');
subplot(3, 4, 5); plot(f,
Coupling, 'b', 'LineWidth', 1.5);
hold on; yline(Ctarget, 'r--', '10
dB Target'); grid on;
title('Coupling');
```

```
xlabel('Frequency (GHz)');
ylabel('Coupling (dB)');

subplot(3, 4, 6); plot(f, Isolation, 'b',
'LineWidth', 1.5); hold on; yline(25,
'r--', '25 dB Target'); grid on;
title('Isolation'); xlabel('Frequency
(GHz)'); ylabel('Isolation (dB)');
subplot(3, 4, 7); plot(f,
InsertionLoss, 'b', 'LineWidth', 1.5);
hold on; yline(1.5, 'r--', '1.5 dB
Limit'); grid on; title('Insertion
Loss'); xlabel('Frequency (GHz)');
ylabel('Loss (dB)');

subplot(3, 4, 8); plot(f, Directivity,
'b', 'LineWidth', 1.5); hold on;
yline(15, 'r--', '15 dB Target'); grid
on; title('Directivity');
xlabel('Frequency (GHz)');
ylabel('Directivity (dB)');
subplot(3, 4, 9);
plot(f, S11_div, 'LineWidth',
1.5); grid on;
title('Divider S11');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 10);
plot(f, S21_div, 'b', f, S31_div, 'r--', 'LineWidth',
1.5); hold on;
yline(-3.01, 'k:', 'Ideal -3.01
dB'); grid on;
```

```
title('Divider Output S21
and S31'); xlabel('Frequency
(GHz)'); ylabel('Magnitude
(dB)');
legend('S21', 'S31', 'Location', 'best');

subplot(3, 4, 11);
plot(f, S23_div, 'LineWidth',
1.5); grid on;
title('Divider Output
Isolation');
xlabel('Frequency (GHz)');
ylabel('Magnitude (dB)');

subplot(3, 4, 12);
plot(f, PowerRatio,
'LineWidth', 1.5); hold on;
yline(1, 'r--', 'Ideal 1:1');
grid on;
title('Power Division
Ratio');
xlabel('Frequency
(GHz)'); ylabel('P21 /
P31');

sgtitle('Microwave Directional Coupler and Two-Way Power Divider');
```

Output



5. Calculated Results (From MATLAB Code)

Parameter at 10 GHz	Before optimization	After optimization	Target / interpretation
Coupling	10.00 dB	10.00 dB	Target: 10 dB met in model
Isolation	30.00 dB	31.00 dB	met in model
Insertion loss	0.50 dB	0.40 dB	met in model
Directivity	20.00 dB	21.00 dB	met in model

6. Output Parameters

Output parameter	Value at 10 GHz	Interpretation
Divider S21	3.01 dB	Output 1 receives approximately half the input power in the ideal equal-split model.

Divider S31	3.01 dB	Output 2 receives approximately half the input power in the ideal equal-split model.
Power division ratio	1.000:1	Equal power division.
Output-port isolation S23	30.00 dB	Illustrative output-port isolation curve.
Input reflection S11 (coupler)	25.00 dB	Illustrative input matching at the design frequency.